NANYANG TECHNOLOGICAL UNIVERSITY

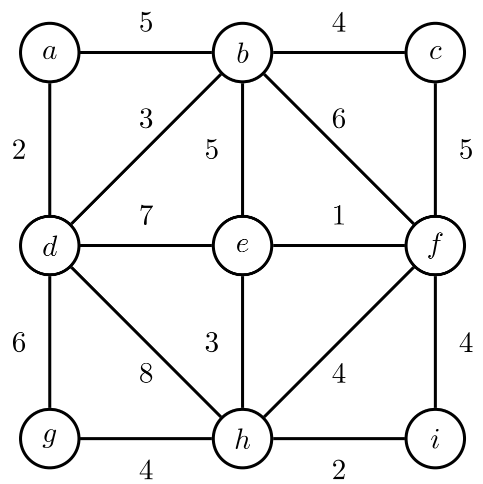
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2 MH1301 Discrete Mathematics Tutorial 12

For the tutorial in Week 13 on 18 April, let us discuss the following questions.

Questions 1 and 2. (Ex.11.5.2, 11.5.4.)

1. Use Prim's algorithm to find a minimum spanning tree for the given weighted graph.
2. Use Kruskal's algorithm to find a minimum spanning tree for the given weighted graph.



Solution:

1. Since the starting vertex is not stated, we shall just start with vertex a. By Prim's algorithm, the edges added to the MST are

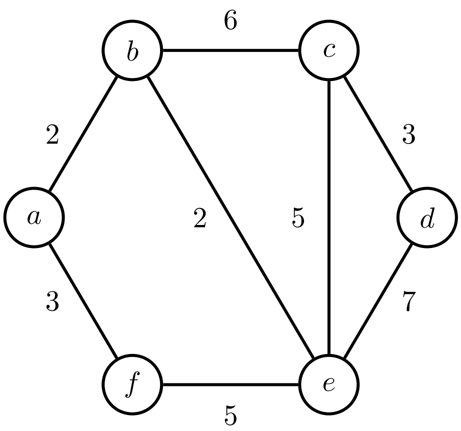
$$\{a,d\},\{b,d\},\{b,c\},\{b,e\},\{e,f\},\{e,h\},\{h,i\},\{g,h\}.$$

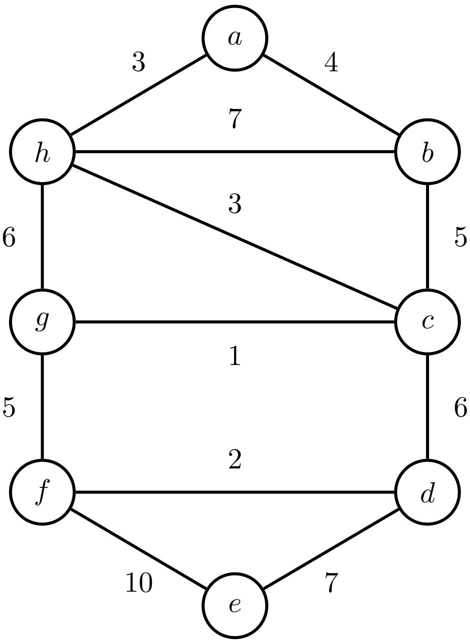
2. By Kruskal's algorithm, the edges are added to the MST in the order

$$\{e,f\},\{a,d\},\{h,i\},\{b,d\},\{e,h\},\{b,c\},\{g,h\},\{b,e\}.$$

Note that the MST is not unique, since the edge $\{b,e\}$ could have been replaced with $\{c,f\}$.

Question 3. Find a minimum spanning tree for each of the following two graphs. Use a different algorithm for each graph. For each graph, state the name of the algorithm used, and list the edges in the order that they are added to the MST.





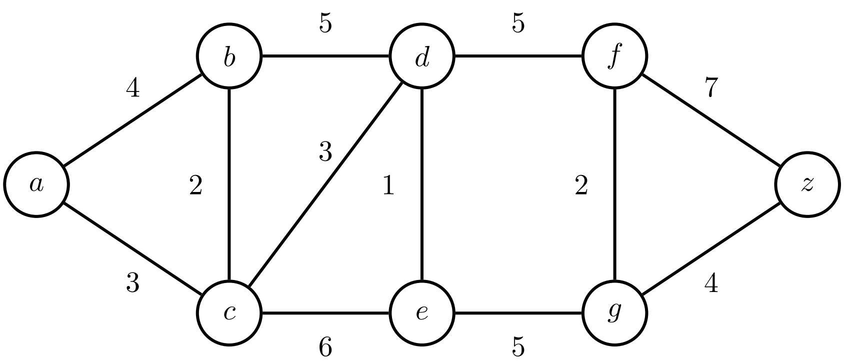
Solution: We apply Prim's algorithm to the first graph. Starting from vertex a, the edges added to the MST are

$$\{a, b\}, \{b, e\}, \{a, f\}, \{e, c\}, \{c, d\}.$$

We apply Kruskal's algorithm to the second graph. The edges added to the MST are

$$\{g, c\}, \{f, d\}, \{a, h\}, \{h, c\}, \{a, b\}, \{g, f\}, \{d, e\}.$$

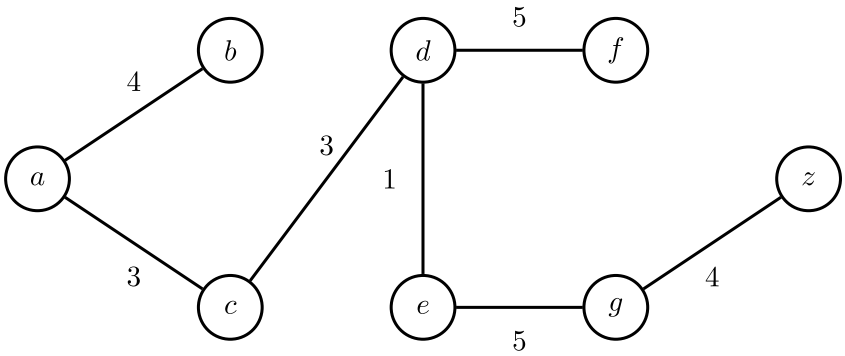
Question 4. (Ex. 10.6.2.) Find the distance between a and z in the following graph using Dijkstra algorithm.
Illustrate how the values of $d[v]$ are updated using a table.



Solution: Here is a table illustrating how the $d[v]$ are updated.

$d[a]$	$d[b]$	$d[c]$	$d[d]$	$d[e]$	$d[f]$	$d[g]$	$d[z]$	S
0	∞	∞	∞	∞	∞	∞	∞	$\emptyset$
0	4	3	∞	∞	∞	∞	∞	$\{a\}$
0	4	3	6	9	∞	∞	∞	$\{a, c\}$
0	4	3	6	9	∞	∞	∞	$\{a, c, b\}$
0	4	3	6	7	11	∞	∞	$\{a, c, b, d\}$
0	4	3	6	7	11	12	∞	$\{a, c, b, d, e\}$
0	4	3	6	7	11	12	18	$\{a, c, b, d, e, f\}$
0	4	3	6	7	11	12	16	$\{a, c, b, d, e, f, g\}$
0	4	3	6	7	11	12	16	$\{a, c, b, d, e, f, g, z\}$

Here is the tree giving the shortest path from vertex a to the other vertices.



From the tree, we see the distance between a and z is 16 via the path $a - c - d - e - g - z$.

Question 5. (Ex 10.6.4). Find the length of a shortest path between these pairs of vertices in the weighted graph in Question 4.

(a) a and d
(b) a and f
(c) c and f
(d) b and z

Solution:

(a) From question 4, the shortest path from a to z is 16 via the path $a - c - d - e - g - z$. Therefore, by lemma from lecture, the shortest path from a to d is via $a - c - d$ and has length 6.
(b) Again, from question 4, the shortest path from a to f is $a - c - d - f$ and has length 11.
(c) Since the shortest path from a to f is $a - c - d - f$, the shortest path from c to f is $c - d - f$ and has length 8. (Otherwise, suppose there is a short path from c to f, say $c - v_1 - \cdots - v_k - f$. Then $a - c - v_1 - \cdots - v_k - f$ is a shorter part than $a - c - d - f$, which is a contradiction.)
(d) Note that b is adjacent to a, c, and d. Therefore, it suffices to compare the $b - a - \cdots - z$, $b - c - \cdots - z$, and $b - d - \cdots - z$.

We see that the shortest paths are given by $b - d - e - g - z$ and $b - c - d - e - g - z$, both having length 15.

Questions 6 and 7. One of the basic motivations behind the Minimum Spanning Tree Problem is the goal of designing a spanning network for a set of nodes with minimum total cost. Here we explore another type of objective: designing a spanning network for which the most expensive edge is as cheap as possible.

Specifically, let $G = (V, E)$ be a connected graph with n vertices, m edges, and positive edge costs that you may assume are all distinct. Let $T = (V, E')$ be a spanning tree of G; we define the *bottleneck edge* of T to be the edge of T with the greatest cost.

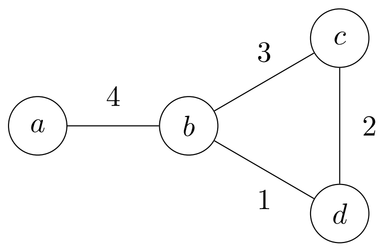
A spanning tree T of G is a *minimum bottleneck spanning tree* if there is no spanning tree T' of G with a cheaper bottleneck edge.

6. Is every minimum-bottleneck tree of G a minimum spanning tree of G? Prove or give a counterexample.

7. Is every minimum spanning tree of G a minimum-bottleneck tree of G? Prove or give a counterexample. (**Hint:** use the property of safe edges.)

Solution:

6. This claim is false. Consider the following graph:



The spanning tree with edges $\{\{a, b\}, \{b, c\}, \{c, d\}\}$ is a minimum-bottleneck spanning tree, but not a minimum spanning tree of G.

7. This claim is true. We prove this by contradiction. Suppose not, that is, suppose there exists a graph $G = (V, E)$ and a MST T of G, such that T is not a minimum bottleneck spanning tree. Let $e = \{u, v\}$ be the bottleneck edge of T. By assumption we know that there exists another spanning tree T', such that for every edge $e' \in T'$, we have $w(e) > w(e')$.

Now, let V_1 be the subset of vertices of V that are reachable from u in T, without going though v. Define V_2 similarly to be the subset of vertices that are reachable from v in T without going through u. By the safe edge theorem, $e = \{u, v\}$ should be a safe edge of cut (V_1, V_2). That is, there are no edges across this cut (V_1, V_2) of weight less than $w(e)$.

However, this is a contradiction because the minimum bottleneck spanning tree T', being a spanning tree, must have an edge across this cut.

Question 8. Let the complete graph K_n have vertices $\{1, 2, \ldots, n\}$ and suppose that for each $u, v \in \{1, 2, \ldots, n\}$, the edge $\{u, v\}$ has weight $w(\{u, v\}) = u + v$. For example, edge $\{1, 2\}$ will have weight $w(\{1, 2\}) = 1 + 2 = 3$, edge $\{2, 4\}$ will have weight $w(\{2, 4\}) = 2 + 4 = 6$, etc.

Determine the minimum spanning tree of this graph, and *compute* the total cost for this minimum spanning tree.

(**Hint:** apply Kruskal's algorithm to this graph, and see what are the edges added to the MST.)

Solution: We prove that for every n, Kruskal's algorithm will choose edges $\{1,2\},\{1,3\},\ldots,\{1,n\}$. Hence, the total cost of this minimum spanning tree is

$$3+4+\cdots+n+(n+1)=\frac{1}{2}(n+1)(n+2)-3.$$

We order the edges by their weights in increasing order. We claim that for each set of edges with the same weight k (i.e., these are the edges $\{1,k-1\},\{2,k-2,\},\ldots$), Kruskal's algorithm will always choose the edge $\{1,k-1\}$. We prove this by induction.

Base case: This is certainly true for $\{1,2\}$ and $\{1,3\}$.

Inductive step: For edge weight $k\geq 5$, let's suppose that the claim is true for $1,\ldots,k-1$, that is, Kruskal's algorithm selects edges $\{1,2\},\{1,3\},\ldots,\{1,k-2\}$, and reject all others, then it is going to consider the edges $\{1,k-1\},\{2,k-2\},\ldots$. Since edge $\{1,k-1\}$ together with previous selected edges form a tree, Kruskal's algorithm will accept it. The remaining edges with weight k are i,j, where $2\leq i,j\leq k-2$. Since $\{1,i\}$ and $\{1,j\}$ are already picked by Kruskal's algorithm, adding $\{i,j\}$ will create a cycle $1-i-j-1$. Hence Kruskal's algorithm will reject it.

Therefore, by induction, Kruskal's algorithm will choose edges $\{1,2\},\{1,3\},\ldots,\{1,n\}$.

The algorithm stops after edge $\{1,n\}$ is chosen as it has formed a spanning tree.

Answer.

4) 16.
5) (a) 6; (b) 11; (c) 8; (d) 15.
6) False.
7) True.
8) $\frac{1}{2}(n+1)(n+2)-3$.